

Flask Learning – Clear Notes (Beginner to Intermediate)

◆ Prerequisites to Learn Flask

Before learning Flask, you should know:

1. Python
 2. SQL (Database)
 3. HTML
 4. CSS
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1. Python Basics (Must Know)

✓ Variables & Data Types

- **Variables** → Used to store data
- **Data Types:**
 - `int` → 10
 - `float` → 10.5
 - `str` → "hello"
 - `bool` → True / False

Collection Data Types:

- **List** → `[1, 2, 3]`
 - **Tuple** → `(1, 2, 3)`
 - **Set** → `{1, 2, 3}`
 - **Dictionary** → `{"key": "value"}`
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Type Casting

- Converting one type to another

```
int("10")  
float(5)
```

`str(100)`

Operators

- Arithmetic → `+` `-` `*` `/` `%`
 - Comparison → `==` `!=` `>` `<`
 - Logical → `and` `or` `not`
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Control Statements

◆ Conditional Statements

if condition:

`pass`

elif condition:

`pass`

else:

`pass`

- Nested `if`
 - Multiple conditions
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◆ Loops

for i in range(5):

`print(i)`

while condition:

`pass`

◆ Jumping Statements

- `break` → stop loop
 - `continue` → skip iteration
 - `pass` → do nothing
 - `return` → exit function
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Functions

◆ Predefined Functions

- `print()`, `len()`, etc.

◆ User-defined Functions

```
def my_function():  
    return "Hello"
```

Modules

- Reusable Python files

```
import math
```

List Comprehension

```
[x for x in range(10)]
```

Methods vs Functions

- **Function** → independent
 - **Method** → belongs to object (`str.upper()`)
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2. SQL (Database Basics)

◆ Types of Commands

● DDL (Data Definition Language)

- `CREATE`
 - `ALTER`
 - `DROP`
 - `TRUNCATE`
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DML (Data Manipulation Language)

- INSERT
 - UPDATE
 - DELETE
-

DQL (Data Query Language)

- SELECT
-

TCL (Transaction Control Language)

- COMMIT
 - ROLLBACK
 - SAVEPOINT
-

DCL (Data Control Language)

- GRANT
 - REVOKE
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Important SQL Concepts

- LIKE operator
 - LIMIT, OFFSET
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3. HTML (Frontend Structure)

Basics

- Tags & Elements
- Heading → `<h1>`
- Paragraph → `<p>`
- Lists → `<ul>`, `<ol>`

- Forms → `<form>`
 - Tables → `<table>`
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Important Tags

- Anchor → `<a>`
 - Image → `<img>`
 - Horizontal line → `<hr>`
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4. CSS (Styling)

◆ Basics

- Styling web pages

◆ Types:

- Inline CSS
 - Internal CSS
 - External CSS
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◆ Selectors

- `class`
 - `id`
-

◆ Basic Styling

- Colors
 - Margin / Padding
 - Fonts
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Full Stack Development Overview

◆ What is Full Stack?

👉 Combination of **Frontend** + **Backend**



Frontend Technologies

1. HTML → Structure
 2. CSS → Styling
 3. Bootstrap → Responsive design
 4. JavaScript → Dynamic behavior
 5. React JS → High performance UI
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Backend Technologies

1. Python
 2. Flask (Framework)
 3. MySQL (Database)
 4. APIs
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Client-Server Architecture

◆ Flow:

1. **Client (Frontend)** sends request
 2. **Server (Backend)** processes request
 3. **Database** stores/retrieves data
 4. Server sends **response** back
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Simple Flow

Client → Request → Server → Database

Client ← Response ← Server



Flask in Real Time

- Flask connects:
 - Frontend (HTML/CSS)
 - Backend (Python logic)
 - Database (MySQL)
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Extra Concepts

◆ Hybrid Jobs

👉 Combination of:

- WFH (Work From Home)
 - WFO (Work From Office)
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Final Learning Path (Correct Order)

1. Python Basics
2. SQL
3. HTML
4. CSS
5. JavaScript
6. Flask
7. Database Integration
8. APIs